

Complete SQL Notes (Beginner to Advanced)

Page 1: 1. Introduction

SQL (Structured Query Language) is used to create, manage and query relational databases.

Example:

```
SELECT * FROM Employee WHERE Salary > 50000;  
CREATE TABLE Student(id INT PRIMARY KEY,name VARCHAR(50));
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Page 2: 2. Database Basics

Tables, rows, columns, primary keys, foreign keys.

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Page 3: 3. DDL

CREATE, ALTER, DROP, TRUNCATE with syntax and examples.

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Page 4: 4. DML

INSERT, UPDATE, DELETE examples.

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INSERT, UPDATE, DELETE examples.

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```

Page 5: 5. DQL

SELECT, WHERE, ORDER BY, LIMIT.

Example:

```
SELECT * FROM Employee WHERE Salary > 50000;  
CREATE TABLE Student(id INT PRIMARY KEY,name VARCHAR(50));SELECT, WHERE, ORDER  
BY, LIMIT.
```

Example:

```
SELECT * FROM Employee WHERE Salary > 50000;  
CREATE TABLE Student(id INT PRIMARY KEY,name VARCHAR(50));
```

Page 6: 6. Functions

COUNT,SUM,AVG,MIN,MAX.

Example:

```
SELECT * FROM Employee WHERE Salary > 50000;  
CREATE TABLE Student(id INT PRIMARY KEY,name  
VARCHAR(50));COUNT,SUM,AVG,MIN,MAX.
```

Example:

```
SELECT * FROM Employee WHERE Salary > 50000;  
CREATE TABLE Student(id INT PRIMARY KEY,name VARCHAR(50));
```

Page 7: 7. GROUP BY

Grouping and HAVING.

Example:

```
SELECT * FROM Employee WHERE Salary > 50000;  
CREATE TABLE Student(id INT PRIMARY KEY,name VARCHAR(50));Grouping and HAVING.
```

Example:

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SELECT * FROM Employee WHERE Salary > 50000;  
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```

Page 8: 8. Joins

INNER, LEFT, RIGHT, FULL, SELF, CROSS.

Example:

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SELECT * FROM Employee WHERE Salary > 50000;  
CREATE TABLE Student(id INT PRIMARY KEY,name VARCHAR(50));INNER, LEFT, RIGHT,  
FULL, SELF, CROSS.
```

Example:

```
SELECT * FROM Employee WHERE Salary > 50000;  
CREATE TABLE Student(id INT PRIMARY KEY,name VARCHAR(50));
```

Page 9: 9. Constraints

NOT NULL, UNIQUE, PRIMARY KEY, FOREIGN KEY, CHECK, DEFAULT.

Example:

```
SELECT * FROM Employee WHERE Salary > 50000;  
CREATE TABLE Student(id INT PRIMARY KEY,name VARCHAR(50));NOT NULL, UNIQUE,  
PRIMARY KEY, FOREIGN KEY, CHECK, DEFAULT.
```

Example:

```
SELECT * FROM Employee WHERE Salary > 50000;  
CREATE TABLE Student(id INT PRIMARY KEY,name VARCHAR(50));
```

Page 10: 10. Views

Creating and using views.

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CREATE TABLE Student(id INT PRIMARY KEY,name VARCHAR(50));
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Creating and using views.

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Page 11: 11. Indexes

Purpose and syntax.

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Purpose and syntax.

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Page 12: 12. Subqueries

Single-row and multi-row.

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SELECT * FROM Employee WHERE Salary > 50000;  
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Single-row and multi-row.

Example:

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Page 13: 13. Set Operators

UNION, UNION ALL, INTERSECT, EXCEPT.

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UNION, UNION ALL,
INTERSECT, EXCEPT.

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Page 14: 14. Transactions

COMMIT, ROLLBACK, SAVEPOINT.

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SELECT * FROM Employee WHERE Salary > 50000;  
CREATE TABLE Student(id INT PRIMARY KEY,name VARCHAR(50));COMMIT, ROLLBACK,  
SAVEPOINT.
```

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Page 15: 15. Normalization

1NF,2NF,3NF,BCNF.

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Page 16: 16. Stored Procedures

Concepts.

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```
SELECT * FROM Employee WHERE Salary > 50000;  
CREATE TABLE Student(id INT PRIMARY KEY,name VARCHAR(50));Concepts.
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Page 17: 17. Triggers

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Page 18: 18. Practice

50 practice ideas and sample queries.

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sample queries.
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Page 19: 19. Interview

Common SQL interview questions and answers.

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Page 20: 20. Cheat Sheet

Summary of commands.

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Page 21: Practice Page 21

Write SQL queries for sample Employee, Student, Orders, and Library databases. Experiment with joins, grouping, subqueries, and transactions.

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